

Bohr's model of the atom

The evidence that made people take the idea of quanta seriously came from the work of the Dane, Niels Bohr. In 1911, Ernest Rutherford, from New Zealand, had discovered that the atom consists of a central, positively charged nucleus surrounded by a cloud of negatively charged electrons. It was natural to think of the electrons as somehow "in orbit" around the nucleus, similar to the way that the planets are in orbit around the Sun. But there is a big difference. According to classical theory, an electrically charged particle orbiting in this way should continuously radiate away electromagnetic energy, and spiral in to the nucleus. What was holding atoms up?

Bohr said that electrons could only occupy certain orbits around the nucleus, and that each one corresponded to a precise amount of energy that was a multiple of the basic quantum. No in-between orbits were allowed, because it was not possible to have a fraction of a quantum (just as, if we push the cash analogy to the limit, there is no physical piece of money smaller than 1¢, and you can't have, say, 12.7¢ in your pocket). So electrons could not radiate energy continuously and spiral inward. What they could



Danish physicist
Niels Bohr
(1885–1962) was
the first to apply
quantum physics

to a description of the atom. Bohr developed his key ideas when he worked with Ernest Rutherford in Manchester, England, between 1912 and 1916, but spent most of his career in Denmark. There, he established the Institute for Theoretical Physics (later known as the Niels Bohr Institute). He was awarded the Nobel Prize in 1922.



invisible force
If electrons were
literally "in orbit"
around atomic
nuclei, they would
radiate energy and
fall inward. What
holds them up?

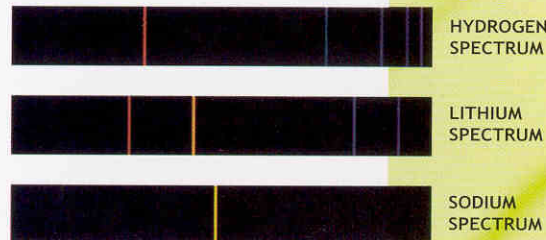
do was jump straight from one orbit to another, either emitting or absorbing a quantum of energy in the process.

The beauty of this idea was that it was already known that atoms of each element produce their own characteristic lines in the spectrum, as distinctive as a bar code. The lines correspond to a precise wavelength, which translates into a precise amount of energy. Bright lines correspond to energy radiated by an atom; dark lines to energy absorbed by an atom. The pattern of those lines made it possible to calculate the energies involved if electrons really were jumping around in this way inside atoms, and the observations matched Bohr's calculations.

The final ingredient in Bohr's model of the atom was his idea that an orbit only has room for a certain number of electrons. If the orbit was full, no more electrons could be added, even if there were atoms in higher orbits that would "like" to make the jump to lower energy levels. So atoms were stable, and the distinctive patterns of lines in the spectrum of light from different elements was explained. Bohr's model of the atom, presented in 1913, made people sit up and take notice of quantum physics as something more than a mathematical trick; but progress in this new branch of physics was delayed by World War I. It was in the 1920s that ideas and conjecture came together in a fully explained quantum theory.



**elemental
bar code**
Just as the bar
code on a can
tells you what
is inside, so the
spectrum of an
element, such as
hydrogen, tells
you which atoms
are present.



**fingerprints
of light**
Three examples
of "pure" spectra
obtained from three
different elements.
Each kind of hot
(excited) atom
produces its own
characteristic
"bar code," so the
elements producing
this light can be
unambiguously
identified.

**"Anyone who is not
locked by quantum theory
is not understood it."**

Niels Bohr